

Ablation studies

Project: Reproducible AI Research Study

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Multiple comparisons

Without opening the old lesson, explain one key idea from Multiple comparisons. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Effect size and practical importance

Without opening the old lesson, explain one key idea from Effect size and practical importance. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Confounders and controls

Without opening the old lesson, explain one key idea from Confounders and controls. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Ablation studies. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Ablation studies. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

Full12/15, without A9/15, without B11/15. Compute sample drops and state interaction limit.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

6. W2 Trace the worked case

Full system8/10, without retrieval5/10, without validation7/10; the last permits an unauthorised action. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

7. W3 Audit the limit

A report omits this qualification: Removing a component changes a complete procedure; attribution depends on the controlled design and possible interactions. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

8. W4 Apply a new case independently

Full12/15, without A9/15, without B11/15. Compute sample drops and state interaction limit.

My reasoning and evidence

9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled ablation studies evidence sheet to Reproducible AI Research Study. Show how you can explain the mechanism and terminology of ablation studies. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Ablation studies

Remove one component at a time. A retrieval pipeline has search plus a relevance filter. A team tests the filter's contribution.

Full pipeline: 8/10 correct. Filter removed: 6/10 on the same cases, with all other settings fixed.

Relevance-filter ablation

correct answers out of 10 shared cases



Bar length shows the supplied value; the marked baseline is zero.

Ablation studies

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Name the component

The relevance filter is the single declared change.



2. Run the ablation

Remove only the filter and preserve cases and scoring.



3. Compare outcomes

Correctness falls from 80% to 60%, two cases in this set.

CHANGE ONE THING

Remove the filter and also change the model version.

Explain the changed result using the matching lesson.

Calculate the filter ablation's two rates and count change. Which single component is altered, and why does also changing the model prevent isolating the filter?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

13. Project portfolio milestone

Project: Reproducible AI Research Study

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Remove one component at a time. A retrieval pipeline has search plus a relevance filter. A team tests the filter's contribution.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Correctness falls from 80% to 60%, two cases in this set.

1. Name the component
2. Run the ablation
3. Compare outcomes

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Compare outcomes Correctness falls from 80% to 60%, two cases in this set. An ablation probes a component by removing or changing it while holding the rest fixed. It supports a scoped comparison, not a claim that the component always helps.

Practice 2: explain the deciding evidence

Changed-case practice: Remove the filter and also change the model version. Which conclusion is supported by the supplied facts?

1. The comparison no longer isolates the filter.
2. This one example guarantees correct results for every future input.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The comparison no longer isolates the filter. Two simultaneous changes provide alternative explanations.

Repair a missing building block

Review Multiple comparisons (ai-7/15). Goal: Explain the mechanism and terminology of multiple comparisons. Short practice: Calculate the six-test cutoff and identify which supplied p-values reject. Why must the team count every planned comparison rather than report only its smallest p-value?. Return to this week and explain what you changed.

Review Effect size and practical importance (ai-7/13). Goal: Explain the mechanism and terminology of effect size and practical importance. Short practice: Calculate the absolute error-rate reduction from 20/1000 to 15/1000. Does it meet the ten-fewer-errors threshold, and what changes when B has eight errors?. Return to this week and explain what you changed.

Review Confounders and controls (ai-7/10). Goal: Explain the mechanism and terminology of confounders and controls. Short practice: Draw the four condition/background combinations. Which variables were coupled before, and why would collecting only healthy-white and spotted-blue again leave the confounding unresolved?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

